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The impact of predation by the girdled goby, *Nesogobius* sp. 1, on abundances of meiofauna and small macrofauna

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Abstract

It is now widely accepted that benthic meiofauna and small macrofauna are an important food source for numerous species of benthic fish. The effect of fish predation on prey populations, however, has been controversial. The girdled goby, *Nesogobius* sp. 1, is a benthic marine fish that is abundant on the mudflats of Swan Bay, Victoria. A short-term (48 h) field caging experiment, where gobies were excluded or included at low and high densities, was used to determine the effect of predation. Predation by gobies was found to significantly affect the abundance of their main prey item, juvenile amphipods. However, this effect only occurred at fish densities higher than those normally found in the environment. There were no significant effects on other prey items, and no significant effect on amphipods at "natural" fish densities. Calculations based on the feeding rates of gobies and the abundances of prey at the study site were consistent with the experimental results. Therefore, we conclude that, in the short term at least, predation by *Nesogobius* sp. 1 is unlikely to influence the abundances of its prey. Calculations based on feeding rates suggest that a long-term effect of predation by gobies on amphipods may be possible depending on the turnover rate of prey populations.

Keywords: Caging; Macrofauna; Meiofauna; *Nesogobius* sp. 1; Population regulation; Predation

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1. Introduction

Meiofauna and small macrofauna are an important food source for higher trophic levels, particularly small fishes and juveniles of larger fishes (Gee, 1989; Coull, 1990). As these fish are generally actively-foraging predators, it seems possible that they may influence the abundances of their prey (Choat & Kingett, 1982). However, the significance of predation remains uncertain and controversial, with studies on the importance of predation in organising communities of small benthic organisms yielding conflicting results (Gee, 1989; Coull, 1990).

Some studies have suggested that the effect of fish predation on meiofaunal and small macrofaunal populations is negligible (Alheit & Scheibel, 1982; Choat & Kingett, 1982; Hicks, 1985; Gee, 1987; Gibbons, 1988; Shaw & Jenkins, 1992), with most of these authors concluding that because prey populations are large, and the number of predatory fish comparatively small, predation is unlikely to cause changes in population sizes (Gee, 1989). Furthermore, many small benthic organisms have life-history strategies, such as continuous reproduction and recruitment, and high growth rates, that help to overcome the effects of predation (Hicks, 1984).

Other authors have demonstrated significant reductions in meiofaunal and small macrofaunal abundance in response to predation (Feller & Kaczynski, 1975; Virnstein, 1977; Sibert, 1979; Bell, 1980; De Morais & Bodiou, 1984; Fitzhugh & Fleeger, 1985; Gee et al., 1985; Smith & Coull, 1987; Ellis & Coull, 1989). These authors have found that although predation alone does not control the distribution and abundance of prey species, predator-prey interactions are dominant factors regulating these animals. In some cases, particular prey species may be controlled by fish predators, while the effect on the meiofaunal community as a whole is negligible (De Morais & Bodiou, 1984; Gee et al., 1985; Ellis & Coull, 1989). An impact on one species is often due to active selection of prey by predators (De Morais & Bodiou, 1984), or due to the fact that some small benthic species have refuges to escape predation (Bell, 1980; Ellis & Coull, 1989).

The girdled goby, *Nesogobius* sp. 1, is a benthic marine fish that is common in sheltered habitats of Victoria, Tasmania and South Australia (Edgar et al., 1982; Last et al., 1983). The girdled goby is abundant on mudflats in Swan Bay, Victoria. The diet of *Nesogobius* sp. 1 is dominated by benthic meiofauna, such as gammaridean amphipods, harpacticoid copepods, ostracods and gastropods (Edgar et al., 1982; Robertson & White, 1986). The aim of the present study was to use a short-term manipulative field experiment to determine whether predation by *Nesogobius* sp. 1 is a significant factor influencing meiofaunal and small macrofaunal populations on the mudflats of Swan Bay. This was accomplished by manipulating densities of *Nesogobius* sp. 1 enclosed in cages and comparing prey densities over time in cage treatments and an uncaged control treatment. The results of the caging experiment were compared with field estimates of predatory impact based on abundances and estimated feeding rates of gobies, and abundances of meiofauna and small macrofauna in the environment.

2. Methods

2.1. Study area

The study was conducted in Swan Bay (lat. 38°16.1′, long. 144°39.3′), a small (10 × 3 km), shallow (<3 m deep) temperate zone marine bay on the Bellarine Peninsula, Victoria, Australia. Swan Bay is characterised by large intertidal mudflats with beds of the seagrass *Zostera muelleri* Irmisch ex Aschers, and subtidal beds of *Heterozostera tasmanica* (Martens ex Aschers). The site of the caging experiment was an unvegetated intertidal mudflat adjacent to the township of Queenscliff, at the southern end of Swan Bay. The tidal range in Swan Bay is small (<1 m).

2.2. Field methods

2.2.1. Fish density

The abundance of *Nesogobius* sp. 1 was investigated by daylight sampling over a 48-h period, concurrently with the caging experiment, to determine the "natural" density of this species in the study area. Twenty-four randomly-positioned samples were collected with a 10 × 2 m seine net with a 4 mm mesh. Each set of the seine net sampled a circular area of ≈12 m^2.

As the capture efficiency of the seine net could not be assumed to be 100%, the net efficiency needed to be calibrated. The net efficiency was determined by setting the net in a circle surrounding an area of 12 m^2, into which 20 fish were placed. After 5 min the fish were collected and counted, and thus the capture rate was determined. The experiment was replicated three times. The mean abundance of fish in the environment was determined at the time of calibration, in order to determine the average number of fish already present in the 12 m^2 areas during each calibration trial.

2.2.2. Caging experiment

The cages, measuring 1 × 1 m × 0.4 m high, were constructed using 3 mm PVC sheet. They had no top or base, the substrate forming the base of the cage. Four 14 cm diameter holes were cut in the upper half of each side. The holes, together with the top of the cage, were covered in 1 mm mesh, which was attached to the cages with glue. The mesh on the tops of the cages had a 30 cm slit which was sealed with Velcro and allowed for easy access. The 1 mm mesh size was chosen because it was necessary to enclose gobies while excluding all other predators. The construction of the cages was expected to allow some immigration and emigration of meiofauna and small macrofauna.

The cages were randomly positioned within an area of ≈400 m^2 of mudflat, and were hammered into the sediment to a depth of ≈15 cm. Before hammering the

cages into place, the areas were checked to make sure there were no predators present, such as burrowing crabs, that may also feed on the meiofauna.

The experiment commenced 1 h before low tide on 30 June and was terminated after 48 h, on 2 July, 1993, at the same stage in the tidal cycle. Although the position of the cages was intertidal, the tidal range was moderate during the experiment, so that the cages retained at least 10 cm of water at low tide, and were completely covered with water only for a short time at high tide. There were four treatments and each had four replicates. Fish were added to three cage treatments at densities of 0, 2 and 10 fish per m^2. Uncaged areas of 1 m^2 were used as a fourth treatment for comparison with caged treatments. Sediment cores were used to sample the benthos. Cores were removed with a 32 mm diameter PVC pipe to a depth of 50 mm. The cores had to be of sufficient depth to account for the previous observation of burying behaviour by the gobies. All sampling was conducted at least 15 cm from the cage edge to avoid potential effects of cage structure (Bell & Woodin, 1984). Four randomly-positioned cores were taken at the start and the end of the experiment, at the same stage of the tidal cycle. Comparison of the meiofaunal and small macrofaunal abundance between the beginning and the end of the experiment reduced the influence of high between-cage variability in abundance detected in pilot studies. The mean abundance of organisms taken from the replicate cores from the uncaged areas was also used as a measure of the abundance of small benthic organisms at the study site for calculations of predatory impact. The sediment cores were preserved in 100% ethanol with rose bengal added to stain biological tissue for ease of sorting in the laboratory.

Gobies, 40–60 mm in total length, were collected by seine netting just prior to the start of the experiment, and were held for a few minutes in an aerated bucket. Handling of the fish was minimal and they were added to the cages as the tide started to flood. During the experiment the cages were checked at low tides to ensure that the fish had remained in the cages, that no fish were dead, and that no unwanted predators were present. Cages were also checked for any sediment erosion or deposition around the cage and fouling of the mesh. At the end of the experiment, just after the core samples were taken, the cages were swept repeatedly with a hand net to catch the fish. Gobies were anaesthetised in MS222 and preserved in 100% ethanol for later gut analysis.

2.2.3. Fish diet

Gobies were caught by seine netting from the area adjacent to the caging area just prior to the end of the experiment on 2 July, anaesthetised in MS222 and preserved in 100% ethanol. The gut contents of these fish were used to compare the diet of gobies from the natural environment to that of gobies from the high-density fish treatment. In addition, the gut contents of these fish were used to establish the feeding rate of the gobies, with reference to a published gastric evacuation rate of 0.16 for *Nesogobius* sp. 1 (Robertson & White, 1986). This value was used to calculate the feeding rate of the goby (by multiplying the gastric

evacuation rate by the mean number of prey in the gut contents) for each of the common prey items in the diet. This species is known to feed continuously over 24 h (Robertson & White, 1986). By extrapolating from the estimated feeding rate, the number of each prey taxon that would be consumed each day was determined, and by comparing this number with the abundance of the particular taxa in the field, the percentage of each prey taxon that would be consumed per day was estimated.

2.3. Laboratory methods

2.3.1. Core samples

The core samples were washed through two sieves, 2 and 0.125 mm, which spanned the prey sizes taken by the gobies. The top sieve (2 mm) removed the larger animals and debris, while the bottom sieve (0.125 mm) collected all of the possible prey items, allowing the smaller animals and silt to pass through. The fauna in the 0.125 mm sieve were identified to major taxon and counted.

2.3.2. Fish

The standard length of each fish was measured to the nearest 1 mm. A size range of 40–60 mm was selected from the wild fish, so that the sample would be comparable to the caged fish. Guts were dissected from the gobies and the contents were identified to major taxon and counted.

2.4. Data analysis

The statistical analysis of the caging experiment data was in the form of a split-plot design using the Systat computer program (Wilkinson, 1992). For each of the meiobenthic taxa the design tested for significant differences in abundance between treatments, between times, the interaction between time and treatments, between cages within treatments and the interaction between cages within treatments and time.

Prior to analysis, the assumptions of analysis of variance were tested by examining box plots, the distribution of residuals in relation to model estimates, and the distribution of residuals on a normal probability plot. For all taxa it was not necessary to transform the data to fulfil assumptions of normality and homogeneity of variance. Because multiple ANOVAs were performed on individual meiobenthic taxa, alpha was set at 0.05 divided by the number of tests performed.

The data from the diet comparison between caged (high fish density) and uncaged (natural environment) fish was analysed using t-tests. Data from the composition of the diet of fish from the low fish-density treatment were not analysed due to the small sample size.

3. Results

3.1. Fish density

The mean number of *Nesogobius* sp. 1 in the field estimated from the five net settings taken concurrently with net efficiency trials was 9 fish per 12 m^2 (0.75 fish per m^2). This suggests that the average number of fish retrieved in net efficiency trials if the net were 100% efficient would be 29. The mean number of fish actually retrieved in efficiency trials was 20.3. Therefore, the net efficiency was ≈0.70. This factor was used to calculate the densities of fishes in replicate seines.

Most of the circular seine nettings contained between 0 and 2.0 fish per m^2 (adjusted for net efficiency) (Fig. 1). Only a very small proportion contained more fish, with only two samples containing between 3 and 4.5 fish per m^2. The average density was 1.5 fish per m^2 and the maximum density was 4.5 fish per m^2.

3.2. Benthic faunal composition

The composition of the benthic fauna at the caging site during the experiment is shown in Table 1. Harpacticoid copepods (44%) and nematodes (40%) dominated and ostracods (8%) were also abundant. Other taxa were much less abundant, together comprising ≈8% of the community.

3.3. Caging experiment

3.3.1. Fish

The cages were checked during the experiment to ensure that the gobies had not escaped. It was extremely difficult to see the gobies, as they were cryptically

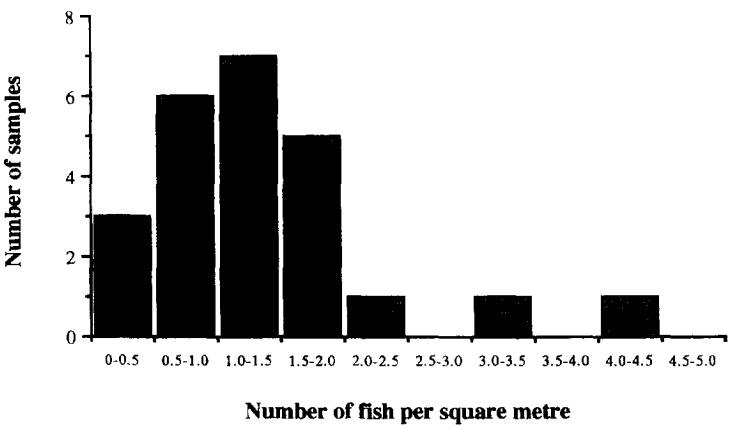

Fig. 1. Abundances of *Nesogobius* sp. 1 at the study site, expressed as the frequency of samples containing ranges of fish densities per square metre.

Table 1
Mean abundance and percentage composition of benthic taxa at the study site

Taxon	Mean abundance (no. $10cm^{-2}$ + SE)	Percentage of Total
Amphipods	7.3 ± 0.4	1.8
Bivalves	9.6 ± 0.5	2.4
Copepods	175.2 ± 5.2	44.3
Gastropods	3.2 ± 0.2	0.8
Isopods	1.6 ± 0.4	0.4
Nematodes	157.9 ± 6.6	39.9
Ostracods	32.3 ± 1.2	8.2
Polychaetes	8.8 ± 0.4	2.2

coloured and often burrowed into the sediment. However, many of the fish were seen after close examination. No dead fish were found in the cages. At the end of the experiment, about half of the fish were recaptured from each cage. As there was only a short period of time available to attempt to net the gobies, we regarded this as a reasonable proportion to recapture. Because no dead fish were collected and no substantial erosion had occurred at the cage edges, we have assumed that the fish placed in the cages had remained until the end of the experiment.

3.3.2. Benthic fauna

The critical term in the analysis of variance was the interaction between treatment and time. A significant decrease in benthic fauna from fish inclusion cages relative to the fish exclusion cage would indicate a significant effect of fish predation.

The only taxon that showed a significant time by treatment interaction in abundance was amphipods (Table 2). This was due to a major (30%) decline in amphipod abundance over the course of the experiment in the high-density fish treatment (Fig. 2). The high-density fish treatment was found to be significantly different from other treatments when the differences in mean abundance in cages between the beginning and end of the experiment were subjected to a one way ANOVA (df = 3/12, $F = 30.73$, $p < 0.05$) and compared using Tukey's test ($p < 0.05$). The similarity between abundances in the exclusion and uncaged treatments suggests little effect of caging artefacts. The lack of significant time by treatment interactions for other taxa suggests that caging and/or predation effects on these species were not important (Table 2). Significant differences between cages within treatments were detected for all taxa except amphipods and gastropods (Table 2), suggesting that for most groups, variability in abundance on the scale of metres between cages was greater than that between replicates within cages. The lack of interactions between time and cages within treatments indicates that differences between cages were constant over time. The results for total benthic fauna (Table 2) reflected the results for the dominant taxa such as nematodes and harpacticoid copepods, with variation between cages the only significant factor.

Table 2
Split plot analysis of variance of caging experiment data for each of the benthic taxa

Source	df	MS	F	MS	F
		Amphipods		Bivalves	
Treatment	3	8.445	5.680 NS	1.083	0.160 NS
Time	1	5.695	32.642*	0.031	0.176 NS
Time * Treatment	3	5.362	30.731*	0.031	0.176 NS
Cage(Treatment)	12	1.487	1.410 NS	6.781	4.227*
Time * Cage(Treatment)	12	0.174	0.165 NS	0.177	0.110 NS
Error	96	1.055		1.604	
		Gastropods		Isopods	
Treatment	3	3.635	4.446 NS	7.987	2.448 NS
Time	1	0.000	1.000 NS	0.070	0.600 NS
Time * Treatment	3	0.188	3.273 NS	0.049	0.422 NS
Cage(Treatment)	12	0.818	1.586 NS	3.263	2.562*
Time * Cage(Treatment)	12	0.057	0.111 NS	0.117	0.092
Error	96	0.516		1.273	
		Ostracods		Polychaetes	
Treatment	3	20.115	0.158 NS	3.865	1.039 NS
Time	1	0.031	0.024 NS	0.000	0.000 NS
Time * Treatment	3	1.031	0.805 NS	0.042	0.444 NS
Cage(Treatment)	12	127.573	18.528*	3.719	5.250*
Time * Cage(Treatment)	12	1.281	0.186 NS	0.094	0.132 NS
Error	96	6.885		0.708	
		Copepods		Nematodes	
Treatment	3	1766.437	1.029 NS	3067.070	0.805 NS
Time	1	2.000	0.109 NS	0.195	0.009 NS
Time * Treatment	3	4.729	0.258 NS	4.279	0.201 NS
Cage(Treatment)	12	1717.464	7.130*	3809.237	14.590*
Time * Cage(Treatment)	12	18.318	0.076 NS	21.237	0.081 NS
Error	96	240.885		261.081	
		Total benthic fauna			
Treatment	3	5121.010	0.609 NS		
Time	1	1.531	0.029 NS		
Time * Treatment	3	9.385	0.175 NS		
Cage(Treatment)	12	8402.786	15.741*		
Time * Cage(Treatment)	12	53.526	0.100 NS		
Error	96	533.813			

Alpha = 0.05/9 = 0.006; NS: $p > 0.006$; *: $p < 0.006$.

3.4. Fish diet

The percentage composition of the diet of fish from uncaged (natural environment) and caged (high fish density) areas is shown in Fig. 3. Both the uncaged and caged fish consumed gammaridean amphipods as a major part of their diet (around 70 and 45%, respectively), both supplementing the diet with smaller prey

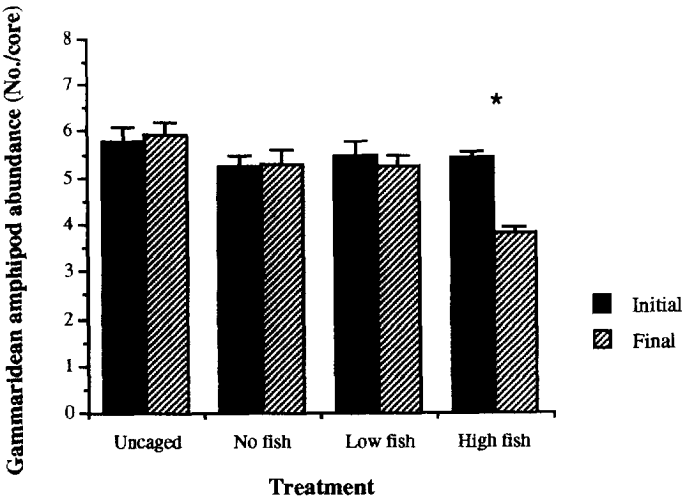

Fig. 2. Mean abundance of gammaridean amphipods in treatments at the beginning and the end of the caging experiment. Error bars represent standard error. Asterix indicates a statistically significant difference.

items such as harpacticoid copepods, ostracods and gastropods. However, the caged fish ate significantly fewer amphipods than the uncaged fish ($t = 2.248$, $df = 37$, $p < 0.05$), and switched to include a significantly higher number of the smaller prey items in their diet ($t = 9.769$, $df = 37$, $p < 0.001$) (Fig. 4). There was no evidence that the feeding of fish had been suppressed by the conditions within the cage.

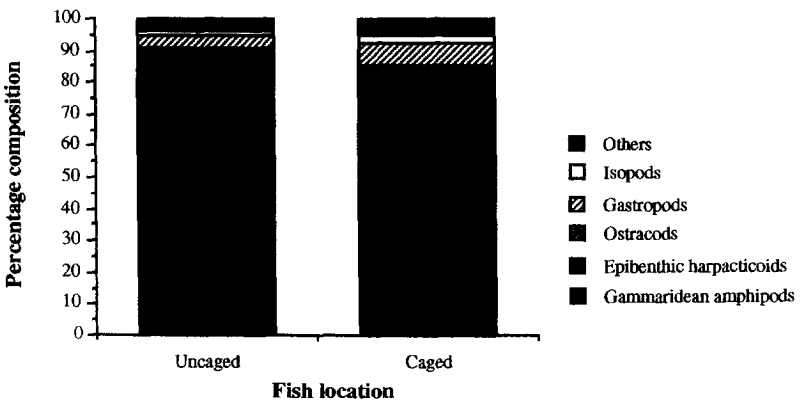

Fig. 3. Percentage composition of the diet of *Nesogobius* sp. 1 from the natural environment (uncaged) and from the high fish-density treatment (caged).

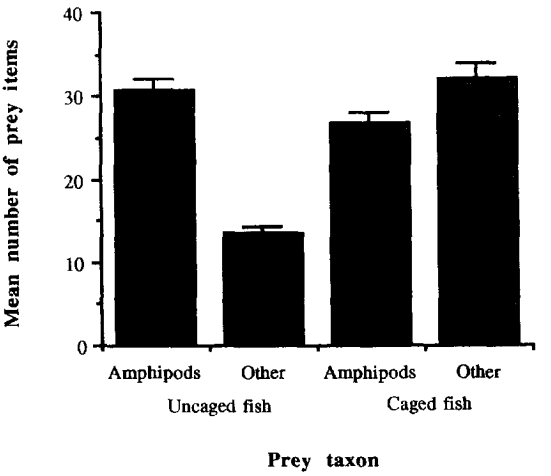

Fig. 4. Numerical composition of the diet of *Nesogobius* sp. 1 from the natural environent (uncaged) and from the high fish-density treatment (caged). The two prey categories are amphipods and other prey items. The "other" items include all other prey taxa taken by the goby, as shown in Fig. 3. Error bars represent standard error.

3.5. Feeding rate

The feeding rate on amphipods was much higher than on other taxa, even though the abundance of amphipods was low compared to other taxa (Table 3). For this reason, the greatest predatory impact occurred on amphipods (Table 3). At an abundance of 2 fish per m^2 only 6% of amphipods would be removed in 48 h (Table 4). However, at a density of 10 fish per m^2, 32% of amphipods would be removed in 48 h. These results are consistent with the results of the caging experiment.

Table 3
Calculations used to determine the percentage of benthic populations consumed each day by *Nesogobius* sp. 1, at a density of 1 fish per m^2, based on data from fish from the natural (uncaged) environment

Prey taxon	Mean prey abundance ($no. \cdot m^{-2}$)	Mean prey number ($no. \cdot fish^{-1}$)	Feeding rate ($no. \cdot hr^{-1}$)	No. of prey consumed ($no. \cdot m^{-2} \cdot day^{-1}$)	Percentage prey removed ($\% \cdot day^{-1}$)
Amphipods	7336.1	30.8	4.93	118.3	1.61
Copepods	175195.1	6.4	1.02	24.5	0.01
Gastropods	3232.8	1.6	0.26	6.1	0.19
Isopods	1616.4	0.5	0.08	1.9	0.12
Ostracods	32204.1	3.5	0.56	13.4	0.04

Table 4
Percentage of benthic faunal populations consumed each day by *Nesogobius* sp. 1, at densities of 1, 1.5, 2, 4.5, and 10 fish per m^2, based on the feeding rate data from Table 3

Prey taxon	Percentage of benthic fauna removed per day				
	1 fish · m^{-2}	1.5 fish · m^{-2}	2 fish · m^{-2}	4.5 fish · m^{-2}	10 fish · m^{-2}
Amphipods	1.61	2.42	3.22	7.25	16.12
Copepods	0.01	0.02	0.02	0.05	0.10
Gastropods	0.19	0.29	0.38	0.86	1.94
Isopods	0.12	0.36	0.24	3.24	1.18
Ostracods	0.04	0.12	0.08	0.18	0.42

4. Discussion

It has been widely suggested that caution must be used in the interpretation of field manipulations of soft sediment benthos, as there are many potential problems associated with these experiments (Virnstein, 1978; Hulberg & Oliver, 1980; Gee et al., 1985; Hall et al., 1990). Artefacts such as erosion and sedimentation, alteration of hydrodynamics, and fouling of meshes may occur when using cages and may influence results. We dealt with caging effects primarily by using predator inclusions for comparison with the predator exclusion treatment, so that any artefacts present would be constant across treatments. Further to this, the lack of significant differences in abundances of faunal groups between caged treatments and the uncaged areas indicated that cage effects were minimal. This was probably due to the short duration of the experiment.

The "natural" fish density treatment approximated the average fish density in the field (1.5 fish per m^2). Since no effect of predation was detected at this density we suggest that none would exist at true field densities. The high fish-density treatment contained a higher number of fish than the maximum density found in the field (4.5 fish per m^2). Thus, the predatory effect found at high fish density is unlikely to be found under normal field conditions. High densities may occur in localised areas over short periods, but averaged over the whole mudflat it is unlikely that there would be an overall effect of predation on amphipod abundance in the study area.

The estimate of a 32.25% reduction of the amphipod population every 48 h if there were 10 fish per m^2, based on the information from feeding rates, was very close to the 30% reduction of amphipods in the caging experiment at high fish density. Such a rate of predation would obviously quickly lead to the decimation of the population. In contrast, feeding rates at the average field density of fish would lead to a reduction of only 2.42% of the amphipod population per day, explaining the lack of significant short-term prey reduction in the experiment. At the maximum fish density (4.5 fish per m^2), 7.25% of the amphipod population may be consumed per day, which would be expected to lead to a significant reduction in prey populations in the short term. Therefore, significant short-term

reductions in prey numbers may occur in localised areas where goby numbers are high. Average feeding rate estimates will be affected by the fact that although this species feeds continuously, feeding intensity shows a tidal rhythm associated with movement over mudflats with the flooding tide (Robertson & White, 1986). This results in a peak in gut contents about 3 h after high tide (Robertson & White, 1986). Our sampling was immediately prior to the low-tide period, and gut contents would be expected to approximate the daily average (Robertson & White, 1986).

The diet of the girdled goby is dominated by gammaridean amphipods, with amphipods comprising around 70% of the diet of fish from the natural environment. As amphipods were much bigger than the other prey taxa, they probably represented an even higher proportion of the diet in terms of energetic return. Thus, gobies showed strong positive selection for amphipods, considering that amphipods numerically comprised only 1.8% of the benthic fauna in the size range considered. For selective predators, the effects on the preferred taxa are likely to be much greater than the effects on minor prey categories (De Morais & Bodiou, 1984).

Although the experiment was not designed to test foraging theory (Pyke et al., 1977), we did note that the diet of the wild fish was slightly different from that of the fish caged at high density. Although both ate predominantly gammaridean amphipods, the caged fish ate significantly fewer of these prey than the uncaged fish, and included significantly more of the smaller items in their diet. Optimal foraging theory suggests that the greatest rewards per unit effort would be obtained by taking the largest handleable prey items (amphipods) (Werner & Hall, 1974; Krebs, 1978). It is possible that less profitable items tended to be ignored when these larger prey items were abundant. However, in the high fish-density treatment, the amphipods may have been depleted due to predation. Thus, the reward obtained per unit effort for amphipods could have declined to below optimum, and the fish switched to include smaller, less profitable items in their diet (Elner & Hughes, 1978; Gibbons, 1988). The increase in consumption of the "other" prey items was much larger than the decrease in consumption of amphipods (Fig. 4). The reason for this is that the "other" prey items are much smaller than amphipods, so relatively more of them must be consumed to compensate for the decrease in amphipod consumption. A comparison by prey volume would probably lead to the same general conclusion, but would show that the increase in the smaller prey items is not as pronounced as it seems in this numerical comparison.

Our conclusion from the experiment that feeding by gobies is relatively unimportant in influencing the abundances of their prey may be challenged on the following grounds. Firstly, benthic fauna may continually migrate to and from the treatment areas, thus masking any effects due to predation (Choat & Kingett, 1982; Kent & Day, 1983; Frid, 1989; Hall et al., 1990). This movement need not be directional, random movement can significantly decrease the effects of prey depletion by a predator (Hall et al., 1990). Although experiments may be confounded, depending on the rates of movement, this is a factor that is not often

addressed. In our study, some mesh was used in the cage design to allow water flow, simulating the natural environment more closely than a completely closed system. Therefore, immigration and emigration of benthic fauna could have occurred. The extent of this migration, however, would have depended on the taxon in question. Nematodes would be unlikely to have burrowed 15 cm to move in or out of cages. It is also unlikely that gastropods and bivalves would undergo any major migration in 2 days. Many other taxa, such as harpacticoids, ostracods, isopods and polychaetes, are known to migrate vertically (Robertson & Howard, 1978; Walters, 1988; Jacoby & Greenwood, 1989; Hall et al., 1990). Thus, it is possible that rapid immigration and emigration of small taxa such as harpacticoids and ostracods, together with young stages of larger taxa such as isopods and polychaetes, may have masked any effects of predation. There is also evidence that some amphipods, depending on species and sex, migrate into the water column and thus could possibly move in and out of cages (Robertson & Howard, 1978; Ambrose, 1984; Jacoby & Greenwood, 1989). However, migration of amphipods was probably more restricted than for copepods and ostracods, as they are much larger, and with the exception of juveniles, would have been restricted by the mesh size to some extent. Although we have not directly measured movements of potential prey, the close correspondence between the result of the caging experiment and that based on feeding rates suggests that immigration and emigration were not significant factors influencing results.

Secondly, some predatory effects in our study may have been masked by the grouping of large numbers of species into broader taxonomic categories (Ellis & Coull, 1989). Vulnerability of individual species would relate to such factors as mobility, visibility, escape responses and depth distribution in the sediment. Whilst predatory effects on individual meiofaunal and small macrofaunal species are certainly possible, the intended focus of our study was on the broader benthic community.

Thirdly, our conclusion that predation has a negligible impact on the abundance of meiofauna and small macrofauna in Swan Bay assumes that *Nesogobius* sp. 1 is the dominant predator on the mudflats. The possible influence of other predators, such as the goby, *Favonigobius lateralis*, juvenile flounder and certain invertebrates, such as shrimp and crabs, cannot be discounted. However, the influence of these other predators is unlikely to affect the abundance of the benthic fauna that is preyed on by *Nesogobius* sp. 1. *Favonigobius lateralis* preys predominantly on harpacticoid copepods (C. Brown & C. Schauble, unpubl. data), and although it is possible that there may be an effect of predation on this prey taxon, it is unlikely considering the high abundance of harpacticoids in the area. It is possible that there is a short-term effect on the meiofauna when juvenile flounder temporarily use the mudflat as a nursery ground (Shaw & Jenkins, 1992). The invertebrate predators occur at a low density, so are unlikely to affect meiofaunal or small macrofaunal abundance. Thus, it seems that the abundances of the taxa that are taken by *Nesogobius* sp. 1 would not be affected to any great extent by the other predators on the mudflat in Swan Bay.

Finally, the short duration of the experiment may not adequately reflect

long-term effects. The duration of the experiment was chosen to minimise caging effects such as scouring and erosion, sedimentation and fouling. At approximately natural densities there was no significant effect of predation over the two days of the experiment. However, the reduction of amphipod populations, uncorrected for tidal variations in prey consumption, of about 2.5% a day may be significant over the longer term depending on the turn-over rate of the population. Production to biomass ratios for amphipods from cool temperate waters are typically about 5 to 10, translating to an increase in population of between 1.4 and 2.8% per day (Edgar, 1990: Table 1). Obviously, amphipod production of this order would be completely removed by gobies. In contrast, production to biomass ratio of a small sub-tropical species was ≈30, suggesting a population increase of 8.2% per day (Fredette & Diaz, 1986). Production of amphipods in Swan Bay of this order would be little affected by goby predation. Actual production of amphipods in Swan Bay is probably somewhere between these extremes. The mudflats in Swan Bay are enriched with organics from seagrass detritus (Shaw & Jenkins, 1992) and water temperatures are relatively mild, averaging ≈16°C. Production of epifauna in Swan Bay is generally high relative to other areas (Bird, 1990). Therefore, a relatively high rate of production of amphipods might be expected.

In conclusion, this study has shown that the abundance of the main prey item of the girdled goby, amphipods, was significantly reduced in cages containing fish at higher than natural density. However, there was no significant effect on amphipods at low fish density, which approximated the average fish density for the environment. No predatory effects on the other prey items were detected at any fish density. Calculations based on the feeding rates of gobies and the prey abundances in the study area were consistent with the experimental results. Our results provide strong evidence that predation by *Nesogobius* sp. 1 is not a major factor influencing meiofaunal and small macrofaunal densities over the short term, however, effects over the longer term will depend on the production rate of amphipods. It would be difficult to design a longer-term experiment that would have minimal caging effects, however, estimates of production rates of amphipods in the area may help clarify the long-term effects of goby predation on this group.

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References

Alheit, J. & A. Scheibel, 1982. Benthic harpacticoids as a food source for fish. *Mar. Biol.*, Vol. 70, pp. 141–147.

Ambrose, W.G. Jr., 1984. Increased emigration of the amphipod *Rhepoxynius atronius* (Barnard) and the polychaete *Nephtys caeca* (Fabricius) in the presence of invertebrate predators. *J. Exp. Mar. Biol. Ecol.*, Vol. 80, pp. 67–75.

Bell, S.S., 1980. Meiofauna-macrofauna interactions in a high salt marsh habitat. *Ecol. Monogr.*, Vol. 50, pp. 487–505.

Bell, S.S. & S.A. Woodin, 1984. Community unity: experimental evidence from meiofauna and macrofauna. *J. Mar. Res.*, Vol. 42, pp. 605–632.

Bird, F., 1990. The relationship between fish predation and abundance, biomass and production of invertebrate prey in seagrass beds in Swan Bay and an adjacent area of Port Phillip Bay, Victoria. Unpubl. Thesis, University of Melbourne, Melbourne, Victoria, 18 pp.

Choat, J.H. & P.D. Kingett, 1982. The influence of fish predation on the abundance cycles of an algal turf invertebrate fauna. *Oecologia*, Vol. 54, pp. 88–95.

Coull, B.C., 1990. Are members of the meiofauna food for higher trophic levels? *Trans. Am. Microsc. Soc.*, Vol. 109, pp. 233–246.

De Morais, L.T. & J.Y. Bodiou, 1984. Predation on meiofauna by juvenile fish in a Western Mediterranean flatfish nursery ground. *Mar. Biol.*, Vol. 82, pp. 209–215.

Edgar, G.J., 1990. The use of size structure of benthic macrofaunal communities to estimate faunal biomass and secondary production. *J. Exp. Mar. Biol. Ecol.*, Vol. 137, pp. 195–214.

Edgar, G.J., P.R. Last & M.W. Wells, 1982. *Coastal fishes of Tasmania and Bass Strait*. Cat & Fiddle Press, Hobart, Tasmania, 176 pp.

Ellis, M.J. & B.C. Coull, 1989. Fish predation on meiobenthos: field experiments with juvenile spot *Leiostomus xanthurus* Lacepede. *J. Exp. Mar. Biol. Ecol.*, Vol. 130, pp. 19–32.

Elner, R.W. & R.N. Hughes, 1978. Energy maximization in the diet of the shore crab, *Carcinus maenas* (L.). *J. Anim. Ecol.*, Vol. 47, pp. 103–116.

Feller, R.J. & V.W. Kaczynski, 1975. Size selective predation by juvenile chum salmon (*Oncorhynchus keta*) on epibenthic prey in the Pugent Sound. *J. Fish. Res. Board Can.*, Vol. 32, pp. 1419–1429.

Fitzhugh, G.R. & J.W. Fleeger, 1985. Goby (Pisces:Gobiidae) interactions with meiofauna and small macrofauna. *Bull. Mar. Sci.*, Vol. 36, pp. 436–444.

Fredette, T.J. & R.J. Diaz, 1986. Life history of *Gammarus mucronatus* Say (Amphipoda: Gammaridae) in warm temperate estuarine habitats, York river, Virginia. *J. Crust. Biol.*, Vol. 6, pp. 57–78.

Frid, C.L.J., 1989. The role of recolonization processes in benthic communities, with special reference to the interpretation of predator-induced effects. *J. Exp. Mar. Biol. Ecol.*, Vol. 126, pp. 163–171.

Gee, J.M., 1987. Impact of epibenthic predators on estuarine intertidal harpacticoid copepod populations. *Mar. Biol.*, Vol. 96, pp. 497–510.

Gee, J.M. 1989. An ecological and economic review of meiofauna as food for fish. *Zool. J. Linn. Soc.* Vol. 96, pp. 243–261.

Gee, J.M., R.M. Warwick, J.T. Davey & C.L. George, 1985. Field experiments on the role of epibenthic predators in determining prey densities in an estuarine mudflat. *Estuarine Coastal Shelf Sci.*, Vol. 21, pp. 429–448.

Gibbons, M.J., 1988. Impact of predation by juvenile *Clinus superciliosus* on phytal meiofauna: are fish important as predators? *Mar. Ecol. Prog. Ser.*, Vol. 45, pp. 13–22.

Hall, S.J., D. Raffaelli & W.R. Turrell, 1990. Predator-caging experiments in marine systems: a re-examination of their value. *Am. Nat.*, Vol. 136, pp. 657–672.

Hicks, G.F.R., 1984. Spatio-temporal dynamics of a meiobenthic copepod and the impact of predation-disturbance. *J. Exp. Mar. Biol. Ecol.*, Vol. 81, pp. 47–72.

Hicks, G.F.R., 1985. Biomass and production estimates for an estuarine meiobenthic copepod, with an instantaneous assessment of exploitation by flatfish predators. *N.Z. J. Ecol.*, Vol. 8, pp. 125–127.

Hulberg, L.W. & J.S. Oliver, 1980. Caging manipulations in marine soft-bottom communities: the importance of animal interactions or sedimentary habitat modifications. *Can. J. Fish. Aquat. Sci.*, Vol. 37, pp. 1130–1139.

Jacoby, C.A. & J.G. Greenwood, 1989. Emergent zooplankton in Moreton Bay, Queensland, Australia: seasonal, lunar, and diel patterns in emergence and distribution with respect to substrata. *Mar. Ecol. Prog. Ser.*, Vol. 51, pp. 131–154.

Kent, A.C. & R.W. Day, 1983. Population dynamics of an infaunal polychaete: the effect of predators and an adult-recruit interaction. *J. Exp. Mar. Biol. Ecol.*, Vol. 73, pp. 185–203.

Krebs, J.R., 1978. Optimal foraging: decision rules for predators. In, *Behavioural ecology, an evolutionary approach*, edited by J.R. Krebs & N.B. Davies, Blackwell, Oxford, pp. 91–121.

Last, P.R., E.O.G. Scott & F.H. Talbot, 1983. *Fishes of Tasmania*. Tasmanian Fisheries Development Authority, Hobart, 563 pp.

Pyke, G.H., H.R. Pulliam & E.L. Charnov, 1977. Optimal foraging: a selective review of theory and tests. *Q. Rev. Biol.*, Vol. 52, pp. 137–152.

Robertson, A.I. & R.K. Howard, 1978. Diel trophic interactions between vertically-migrating zooplankton and their fish predators in an eelgrass community. *Mar. Biol.*, Vol. 48, pp. 207–213.

Robertson, C.H. & R.W.G. White, 1986. Feeding patterns of *Nesogobius* sp., *Gymnapistes marmoratus*, *Neoodax balteatus* and *Acanthaluteres spilomelanurus* from a Tasmanian seagrass meadow. *Aust. J. Mar. Freshwater Res.*, Vol. 37, pp. 481–489.

Shaw, M. & G.P. Jenkins, 1992. Spatial variation in feeding, prey distribution and food limitation of juvenile flounder *Rhombosolea tapirina* Günther. *J. Exp. Mar. Biol. Ecol.*, Vol. 165, pp. 1–21.

Sibert, J.R., 1979. Detritus and juvenile salmon production in the Nanaimo Estuary: 2. Meiofauna available as food to juvenile chum salmon. *J. Fish. Res. Board Can.*, Vol. 36, pp. 497–503.

Smith, L.D. & B.C. Coull, 1987. Juvenile spot (Pisces) and grass shrimp predation on meiobenthos in muddy and sandy substrata. *J. Exp. Mar. Biol. Ecol.*, Vol. 105, pp. 123–136.

Virnstein, R.W., 1977. The importance of predation by crabs and fishes on benthic fauna in Chesapeake Bay. *Ecology*, Vol. 58, pp. 1199–1217.

Virnstein, R.W., 1978. Predator caging experiments in soft sediments: caution advised. In, *Estuarine interactions*, edited by M.L. Wiley, Academic Press, New York, pp. 261–273.

Walters, K., 1988. Diel vertical migration of sediment-associated meiofauna in subtropical sand and seagrass habitats. *J. Exp. Mar. Biol. Ecol.*, Vol. 117, pp. 169–186.

Werner, E.E. & D.J. Hall, 1974. Optimal foraging and the size selection of prey by the bluegill sunfish (*Lepomis macrochirus*). *Ecology*, Vol. 55, pp. 1216–1232.

Wilkinson, L., 1992. *SYSTAT: Statistics, Version 5.2 Edition*. SYSTAT Inc, Evanston, Illinois, 724 pp.